

Timeline Check-in

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1. Set up

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.2.0      v readr      2.2.0  
v forcats    1.0.1      v stringr    1.6.0  
v ggplot2     4.0.2      v tibble     3.3.1  
v lubridate  1.9.5      v tidyr      1.3.2  
v purrr       1.2.1
```

```
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()
```

i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become warnings

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
# file organization
library(here)
```

here() starts at /Users/audreyjuneman/Desktop/UCSB/ENVS/ENVS-193DD/github/group-project-193DD

```
# Boxplot Visualization
```

```
library(hrbrthemes)
```

```
library(viridis)
```

Loading required package: viridisLite

```
# text label instead of legend
```

```
library(cowplot)
```

Attaching package: 'cowplot'

The following object is masked from 'package:lubridate':

stamp

```
# Ridgeline Plot
```

```
#install.packages("ggridges")
```

```
library(ggridges)
```

```
#install.packages("ggplot2")
```

```
library(ggplot2)
```

```
# Color Palette
```

```
library(NatParksPalettes)
```

```
library(wesanderson)
```

Registered S3 method overwritten by 'wesanderson':

method from

print.palette NatParksPalettes

```
# Use ragg_png device for better font rendering and system font support
knitr::opts_chunk$set(dev = "ragg_png")
```

Data

```
# weather data from NOAA weather station
weather <- read_csv(here("data", "NOAA-weather-data.csv"))
```

Rows: 1912 Columns: 9

-- Column specification -----

Delimiter: ","

chr (3): STATION, NAME, DATE

dbl (6): LATITUDE, LONGITUDE, ELEVATION, PRCP, TMAX, TMIN

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# water quality survey metadata
metadata <- read_csv(here("data", "NCOS_YSI_Water_Quality_Monitoring_0.csv"))
```

Rows: 687 Columns: 17

-- Column specification -----

Delimiter: ","

chr (12): GlobalID, Monitoring Date, Monitor Name, Weather, Site Name, Othe...

dbl (4): ObjectID, Water Surface Elevation, x, y

time (1): Time of WQ Measurement

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# water quality parameter measurements
water_parameters <- read_csv(here("data", "YSI_Data_Begin_1.csv"))
```

Rows: 1488 Columns: 13

-- Column specification -----

Delimiter: ","

chr (6): GlobalID, ParentGlobalID, CreationDate, Creator, EditDate, Editor

dbl (7): ObjectID, Depth Code, DO (mg/L), DO (%sat), Conductivity (uS/cm), S...

- i Use ``spec()`` to retrieve the full column specification for this data.
- i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

Cleaning Code

Cleaning Weather

```
weather_clean <- weather |>
  # made column names all lowercase
  clean_names() |>
  # puts date in proper format and makes sure its understood as date (not as chr)
  mutate(date = mdy(date)) |>
  # keeping only these 4 columns of interest
  select(date, prcp, tmax, tmin) |>
  # creating new columns for month and year
  # extracting month from the date (needs to be in the standard date format for this function)
  mutate(month = month(date),
         year = year(date)) |>
  mutate(wy = case_when(
    # when month is october or later, assign a value of year +1
    # in the water year for next year
    month >= 10 ~ year + 1,
    # keeping all months < 10 the same (sept or earlier), assigning value of existing year
    .default = year
  ))
```

Cleaning Metadata

```
metadata_clean <- metadata |>
  # col names easier to work with (all lowercase w/ _)
  clean_names() |>
  # "!" is the "not-operator", columns we DONT want to keep, excluding
  select(!c(object_id, creator, editor)) |>
  mutate(monitring_date = mdy_hms(monitring_date)) |>
  rename(monitring_datetime = monitring_date) |>
  mutate(month = month(monitring_datetime),
         year = year(monitring_datetime)) |>
  # assigning water year
```

```

mutate(wy = case_when(
  month >= 10 ~ year +1,
  .default = year )) |>
# creatig new column for site full name
mutate(site_full = case_when(
  site_name == "east_channel" ~ "East Channel",
  site_name == "phelps_bridge" ~ "Phelps Bridge",
  site_name == "venoco_bridge" ~ "Venoco Bridge",
  site_name == "other" ~ "Other"
)) |>
# filter out Other observations
filter(site_full != "Other")

```

Cleaning Water Parameters

```

# creating a new object from water parameters
water_parameters_clean <- water_parameters |>
# cleaning column names
clean_names() |>
# "!" not operator, excluding unneeded columns
select(!c(creator, editor)) |>
# renaming depth code to be elevation code to better represent methodology
rename(elevation_code = depth_code) |>
## Everything below came from visualizing the data first and defining what to fix/add
# fixing a mistake in the data
mutate(elevation_code = case_when(
  # anytime theres a 10 in elevation code, replace it with a 1
  elevation_code == 10 ~ 1,
  # everything else, keep the same
  .default = elevation_code)) |>
mutate(
  # turning column into a factor
  elevation_code = as_factor(elevation_code),
  # defining the correct order for the levels of the factor
  elevation_code = fct_relevel(elevation_code,
    # write in the order you want
    "0", "1", "2", "3", "4", "5", "6"))

```

Joining Code

Water Quality = Metadata + Water Parameters

```
# full joining
water_quality <- full_join(
  # left data frame
  x = metadata_clean,
  # right data frame
  y = water_parameters_clean,
  # name the columns in each df that you want to join by (naming the shared column)
  by = c("global_id" = "parent_global_id")) |>
select(
  # information about the observation
  global_id, monitoring_datetime, time_of_wq_measurement, monitor_name, weather, site_name
  # water information
  water_surface_elevation, flow, notes,
  # measurement information
  elevation_code, do_mg_l, do_percent_sat, conductivity_u_s_cm, salinity_ppt, temperature_
) |>
# taking the date out of this column, dont want the time
mutate(date = date(monitoring_datetime)) |>
# moving the new date column to a better location within the df
relocate(date, .after = monitoring_datetime) |>
# dont need this column anymore
select(!monitoring_datetime)
```

Water Quality Weather = Water Quality + Weather

```
# left join
water_quality_weather <- left_join(
  # left data frame
  x = water_quality,
  # right data frame
  y = weather_clean,
  # joining these dfs by their date column
  by = "date")
```

Wrangling/Summarizing Code

Salinity Outliers

```
# creating new obj
salinity_outlier_ids <- water_quality_weather |>
  # filtering to only include outlier values
  filter(salinity_ppt > 1410) |>
  # extracting the global id column as a vector using pull
  pull(global_id)
```

Summarizing Precipitation

```
# create new object from `water_quality_weather`
monthly_prcp <- water_quality_weather |>
  # only include elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # get rid of site names w/ "other"
  filter(!site_full == "NA") |>
  # removing any duplicate rows based on the combination of month, year, and wy
  distinct(month, year, prcp) |>
  # grouping by month and year
  group_by(month, year) |>
  # calculating the sum of prcp per month (not per site as it is a weather event and not site)
  summarise(sum_prcp = sum(prcp, na.rm = TRUE)) |>
  # ungrouping
  ungroup() |>
  # creating a date column to be read as a date
  mutate(date = lubridate::make_date(.data$year, .data$month, 1)) |>
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
                        ordered = TRUE))
```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by month and year.
- i Output is grouped by month.
- i Use `summarise(.groups = "drop\_last")` to silence this message.

i Use ``summarise(.by = c(month, year))`` for per-operation grouping (``?dplyr::dplyr_by``) instead.

Filtering and Summarizing Salinity

```
salinity_filtered <- water_quality_weather |>
  # keeping only elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # filtering the column global_id to exclude the values that match those in `salinity_outli
  filter(!global_id %in% salinity_outlier_ids) |>
  # dropping the column for other_site_names
  select(!other_site_name) |>
  # get rid of site names w/ "other"
  filter(!site_name == "other") |>
  # setting the order of the levels (from lowest to highest avg salinity value)
  # helps the order of the visualizations
  mutate(site_full = fct_relevel(site_full,
                                "Phelps Bridge", "East Channel", "Venoco Bridge")) |>

  # taking out sites w/ NA
  filter(!site_full == "NA") |>
  # keeping only columns of interest (to reduce confusion)
  select(site_full, water_surface_elevation, elevation_code, salinity_ppt, temperature_c, mo
  # creating levels to `month`, putting them in a logical order
  mutate(month = factor(month,
                        levels = 1:12,
                        labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "S
                        ordered = TRUE)) |>

  # group by site full, month, and year
  group_by(site_full, month, year) |>
  # calculate mean salinity (per site per month)
  summarize(mean_salinity = mean(salinity_ppt, na.rm = TRUE),
            .groups = "drop")
```

Joining Monthly_prcp and Salinity_filtered

```
salinity_prcp <- salinity_filtered |>
  left_join(monthly_prcp, by = c("month", "year"))
```


Exploratory Figure Code

Exploring Salinity

```
# create new object `exploratory_salinity` from `water_quality_weather`
exploratory_salinity <- water_quality_weather |>
  # only keep elevation codes 0, 1, 2, and 3 (keeping three just for visualization)
  filter(elevation_code %in% c("0", "1", "2", "3")) |>
  # filtering the column global_id to exclude the values that match those in `salinity_outlier_ids`
  filter(!global_id %in% salinity_outlier_ids) |>
  # dropping the column `other_site_names`
  select(!other_site_name) |>
  # filterign out site names that match "other"
  filter(!site_name == "other") |>
  # setting the order of the levels (from lowest tp highest avg salinity value)
  # helps the order of the visualizations
  mutate(site_full = fct_relevel(site_full,
                                "Phelps Bridge", "East Channel", "Venoco Bridge")) |>

# taking out sites w/ NA
filter(!site_full == "NA") |>
# keeping only columns of interest (to reduce confusion)
select(site_full, water_surface_elevation, elevation_code, salinity_ppt, temperature_c, month, year) |>
# group by site, elevation code, month, and year
group_by(site_full, elevation_code, month, year) |>
# calculate mean salinity
summarize(mean_salinity = mean(salinity_ppt, na.rm = TRUE),
           .groups = "drop") |>
# creating levels to `month`, putting them in a logical order
mutate(month = factor(month,
                      levels = 1:12,
                      labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
                      ordered = TRUE))
```

Exploring Water Surface Elevation

```
# create a new object `monthly_surface_elv` from `water_quality_weather`
monthly_surfae_elv <- water_quality_weather |>
  # only include elevation codes 0, 1, and 2
  filter(elevation_code %in% c("0", "1", "2")) |>
  # Filter out site names matchign to "other"
```

```

filter(!site_name == "other") |>
# group by site, month, and year
group_by(site_full, month, year) |>
# calculate mean water surface elevation
summarise(avg_elv = mean(water_surface_elevation, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# create a date column to be read as a date
mutate(date = lubridate::make_date(.data$year, .data$month, 1)) |>
# creating levels to `month`, putting them in a logical order
mutate(month = factor(month,
                      levels = 1:12,
                      labels = c("Jan", "Feb", "Mar", "Apr", "May", "Jun", "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),
                      ordered = TRUE))

```

`summarise()` has regrouped the output.

- i Summaries were computed grouped by site_full, month, and year.
- i Output is grouped by site_full and month.
- i Use `summarise(.groups = "drop\_last")` to silence this message.
- i Use `summarise(.by = c(site\_full, month, year))` for per-operation grouping (`?dplyr::dplyr\_by`) instead.

```

## Joining data frames
elv_viz <- monthly_surfae_elv |>
# joining by columns month, year, and site
left_join(salinity_prctp, by = c("month", "year", "site_full")) |>
# filtering out all NA average elevation values
filter(!avg_elv == "NaN")

```

Creating Precipitation Peaks

```

# creating a new object `precipitation_peaks` from `salinity_prctp`
precipitation_peaks <- salinity_prctp |>
# including the dates of the top 4 monthly precipitation
filter(date %in% c("2021-01-01", "2023-03-01", "2023-02-01", "2021-10-01")) |>
# ungrouping data frame
ungroup() |>
# keeping these 4 distinct dates
distinct(date, .keep_all = TRUE) |>

```


To answer our question, “How does precipitation explain salinity?”, we analysed whether monthly precipitation is related to monthly average salinity across NCOS sites. Our draft code already cleans and joins the weather, metadata, and water-quality datasets, then calculates total monthly precipitation and mean monthly salinity by site.

We also chose to use Spearman's correlation coefficient to test whether salinity tends to decrease as precipitation increases.

The test gave a coefficient value of 0.0166 and p-value of 0.8251, suggesting that precipitation alone does not strongly explain salinity.

[illegible]

Figures

Figure 1. - Monthly Precipitation at NCOS (Time Series w/ Rain Event Lines)

```
# base layer: ggplot
# data: salinity_prcp
ggplot(data = salinity_prcp,
       # x axis: date
       mapping = aes(x = date,
                     # y axis: monthly precipitation sum
                     y = sum_prcp)) +
# prcp displayed by columns (a suggestion made by An after work plan/proposal edits)
geom_col(color = "darkblue") +
# adding themeing
theme_bw() +
# x-axis breaks every 6 months (to reduce cognitive load)
# Format date axis labels as abbreviated month and full year
scale_x_date(date_breaks = "6 months", date_labels = "%b %Y") +
# angling labels to 45 degrees and to be right-aligned
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
# expanding limits to make room for text
scale_y_continuous(limits = c(0, 6)) +
# labeling x and y axes
labs(x = "Date",
     y = "Total Monthly Precipitation",
     # adding title
     title = "Total Monthly Precipitation at NCOS") +
# removing legend, redundant
```

```

theme(legend.position = "none") +
# adding in lines for prcp events
geom_vline(data = precipitation_peaks,
           # centering them on dates
           aes(xintercept = date,
              # coloring them by their established label
              color = label),
           # signified by dotted lines
           linetype = "dotted",
           # width of lines
           linewidth = 1.25) +
# adding custom color palette
scale_color_manual(values = wes_palette("AsteroidCity1")) +
# adding labels to each dotted line from the `precipitation_peaks` object
geom_text(data = precipitation_peaks,
          # x-axis location: date
          # y-axis location: set y value from earlier chunk
          # label = the defined label from earlier chunk
          # color by label
          aes(x = date, y = label_y, label = label, color = label),
          # 45 degree angle for readability
          angle = 45,
          # slightly to right of vertical dotted line
          hjust = -0.1,
          # text size
          size = 3,
          # text bolding
          fontface = "bold")

```

Warning: Removed 3 rows containing missing values or values outside the scale range (``geom_col()``).

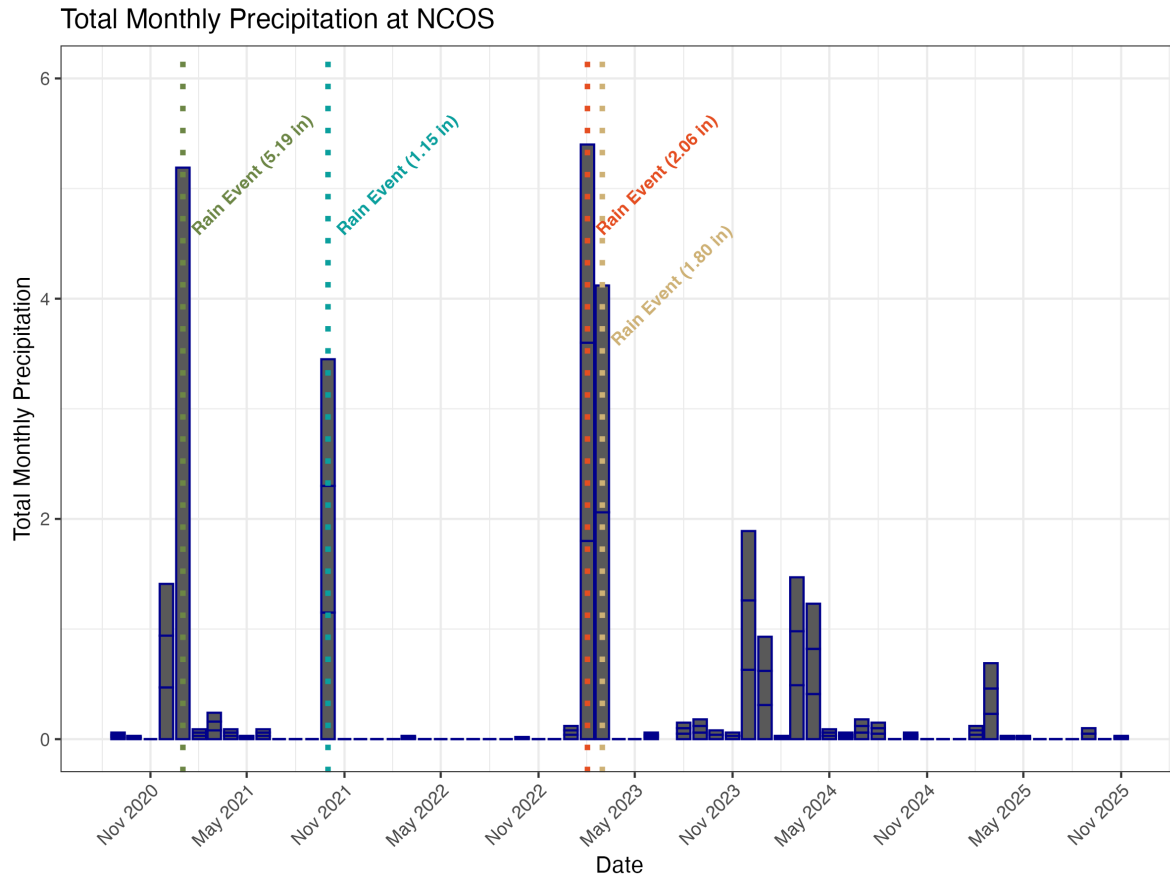


Figure 2. Salinity Time Series w/ Prcp Highlights

```
# base layer ggplot
# data: salinity_prpc
ggplot(data = salinity_prpc,
  # x-axis: date
  mapping = aes(x = date,
    # y-axis: mean_salinity
    y = mean_salinity)) +
  # column displaying mean salinity
  geom_col(color = "grey") +
  # separating points from sites into different plots
  facet_wrap(~ site_full) +
  # adding themeing
  theme_bw() +
```

```

# x breaks by year to reduce cognitive load, labeling by abbreviated month and full year
scale_x_date(date_breaks = "12 months", date_labels = "%b %Y") +
# angling x axis labels to 45 degrees and to be right-aligned
theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
# labeling x and y axes
labs(x = "Date",
      y = "Monthly Average Salinity (ppt)",
      # adding title
      title = "Monthly Average NCOS Salinity",
      # adding legend title
      color = "Precipitation Events") +
# adding lines from the `precipitation_peaks_facet` object
geom_vline(data = precipitation_peaks_facet,
            # oriented on the date
            aes(xintercept = date,
                # different colors by label
                color = label),
            # long dash for better viz
            linetype = "longdash",
            # line width
            linewidth = 0.75) +
# custom colors
scale_color_manual(values = wes_palette("AsteroidCity1")) +
# legend at bottom for less visual clutter
theme(legend.position = "bottom")

```

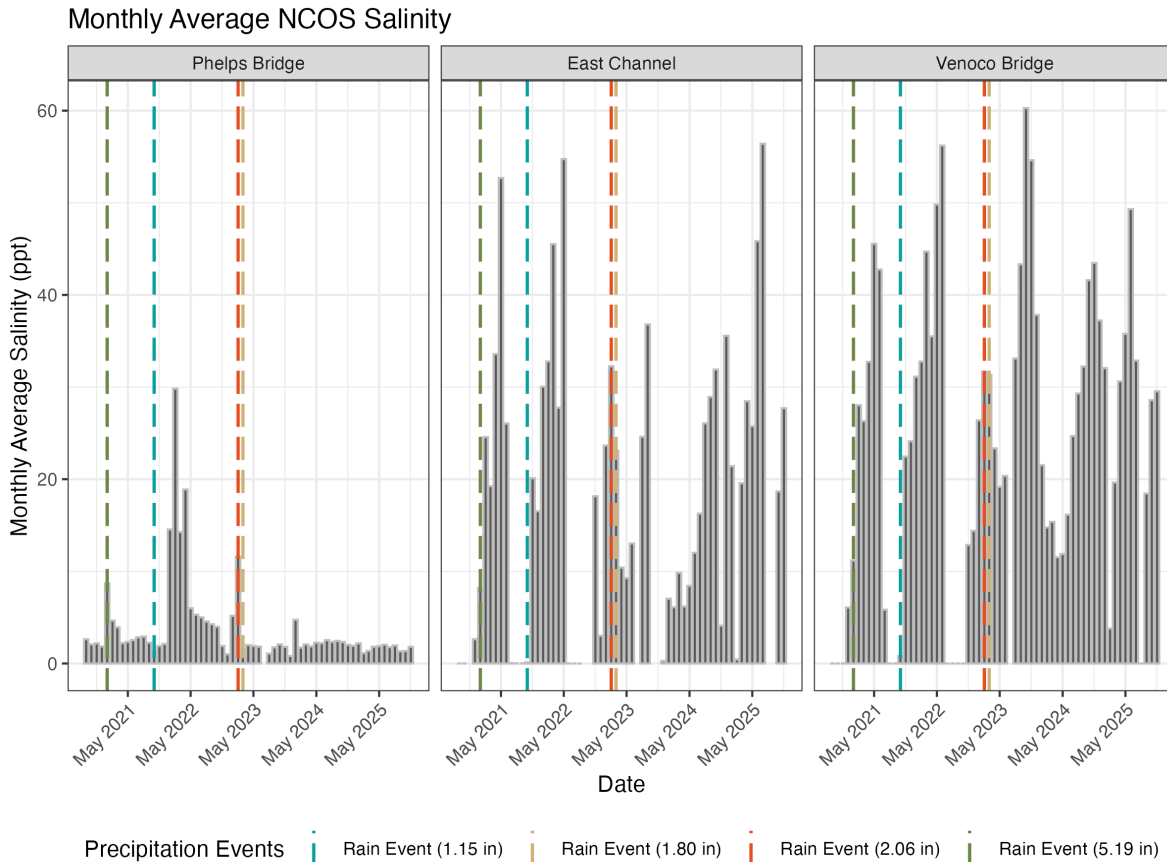


Figure 3. - How Salinity Relates to Water Surface Elevation (Venoco Bridge)

```
ggplot(data = elv_viz,
       mapping = aes(x = avg_elv,
                     y = mean_salinity)) +
  geom_point(color = "maroon") +
  facet_wrap(~ site_full) +
  scale_color_manual(values = wes_palette("GrandBudapest1")) +
  theme_bw() +
  labs(x = "Monthly Average Water Surface Elevation (m)",
       y = "Monthly Average Salinity (ppt)",
       title = "Figure 2. How Salinity Relates to Water Surface Elevation (Venoco Bridge)")
```


Figure 2. How Salinity Relates to Water Surface Elevation (Venoco Bridge)

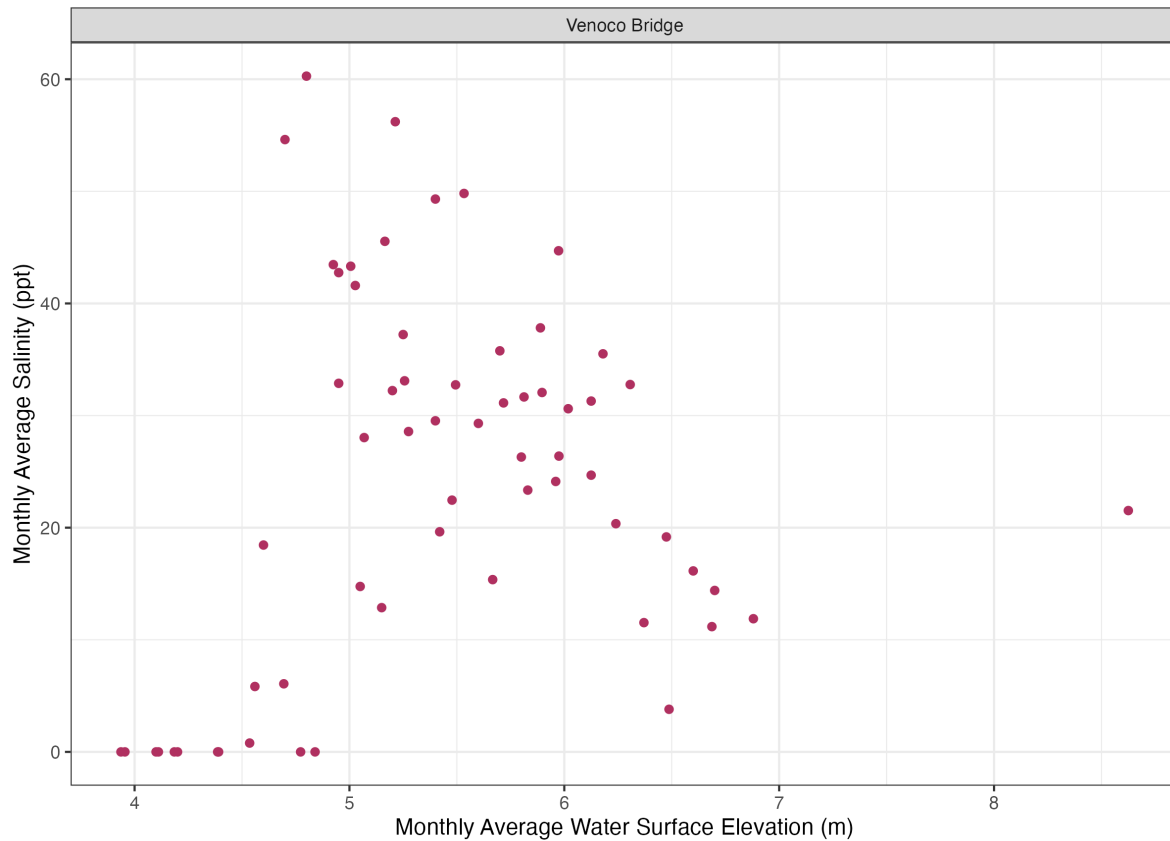


Figure 4. - How Salinity Varies Between Elevation Codes

```
ggplot(data = exploratory_salinity,
       mapping = aes(x = elevation_code,
                     y = mean_salinity)) +
  geom_point(shape = 21) +
  stat_summary(fun = median,
              geom = "point",
              color = "maroon") +
  stat_summary(fun = median,
              geom = "line",
              color = "maroon",
              group = 1) +
  labs(x = "Elevation Code",
       y = "Mean Salinity",
```

```

    title = "Figure 3. Mean Salinity Differences Between Elevation Codes") +
  facet_wrap(~ site_full) +
  theme_bw() +
  scale_x_discrete(breaks = c(0,1, 2, 3)) +
  scale_y_continuous(position = "right") +
  coord_flip()

```

Figure 3. Mean Salinity Differences Between Elevation Codes

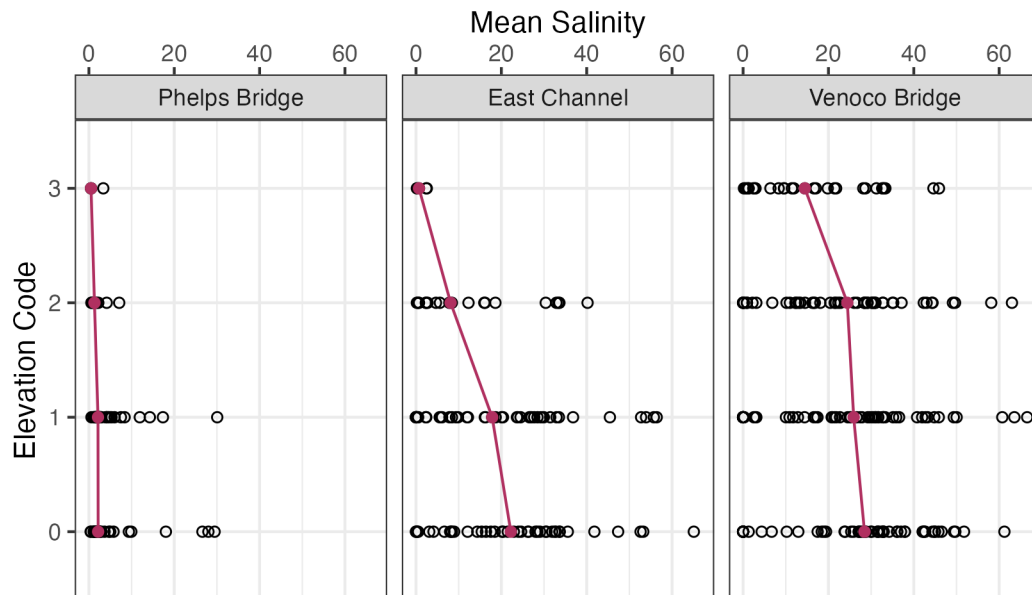


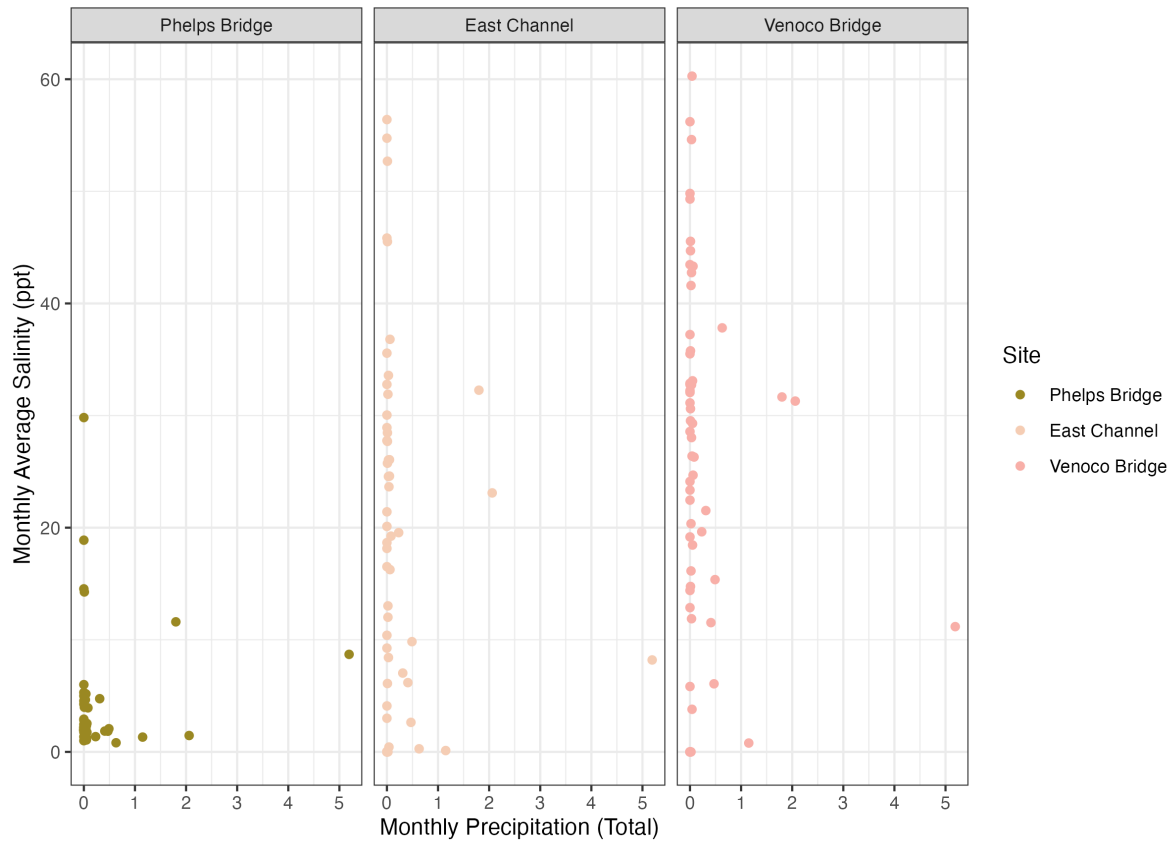
Figure 5. - How Precipitation Explains Salinity

```

ggplot(data = salinity_prctp,
  mapping = aes(x = sum_prctp,
    y = mean_salinity,
    color = site_full)) +
  geom_point() +
  facet_wrap(~ site_full) +
  scale_color_manual(values = wes_palette("Royal2")) +
  theme_bw() +
  labs(x = "Monthly Precipitation (Total)",
    y = "Monthly Average Salinity (ppt)",
    title = "Figure 4. How Precipitation Explains Salinity",
    color = "Site")

```

Figure 4. How Precipitation Explains Salinity



Correlation Test

Calculating Spearman's Correlation

```
cor.test(salinity_prpc$mean_salinity,
         salinity_prpc$sum_prpc,
         method = "spearman")
```

```
Warning in cor.test.default(salinity_prpc$mean_salinity,
salinity_prpc$sum_prpc, : Cannot compute exact p-value with ties
```

```
Spearman's rank correlation rho
```

```
data: salinity_prctp$mean_salinity and salinity_prctp$sum_prctp
S = 939961, p-value = 0.8251
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.01663311
```

Interpreting the Results

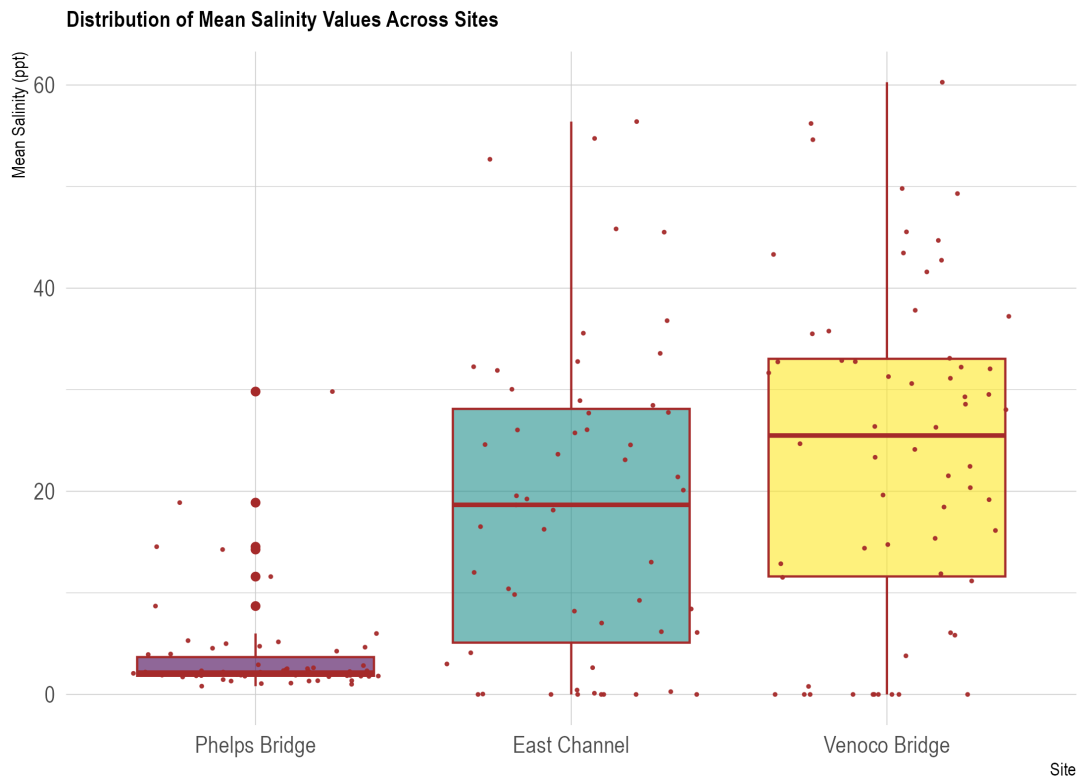
A Spearman's rank correlation was conducted to examine the relationship between monthly precipitation and monthly average salinity. Results indicated no statistically significant relationship between the two variables ($\rho = 0.017$, $S = 939,961$, $p = 0.825$). The near-zero ρ suggests that monthly precipitation has negligible association with average salinity at this scale. It should be noted that exact p-values could not be computed due to ties in the data, so a normal approximation was used; this does not meaningfully affect the interpretation. These results may reflect lag effects between precipitation events and salinity response, spatial or hydrological disconnect between measurement sites, or the influence of other drivers, such as evaporation or tidal exchange, that operate independently of local precipitation.

4. Other exploratory visualizations

Comparing Across Groups

Boxplot w/ Jitter - Distribution of Mean Salinity Values Across Sites

```
# Plot
salinity_prctp |>
  ggplot( aes(x= site_full, y=mean_salinity, fill = site_full)) +
    geom_boxplot(color = "brown") +
    scale_fill_viridis(discrete = TRUE, alpha=0.6) +
    geom_jitter(color="brown", size=0.4, alpha=0.9) +
    theme_ipsum() +
    theme(
      legend.position="none",
      plot.title = element_text(size=11)
    ) +
    ggtitle("Distribution of Mean Salinity Values Across Sites") +
    xlab("Site") +
    ylab("Mean Salinity (ppt)")
```

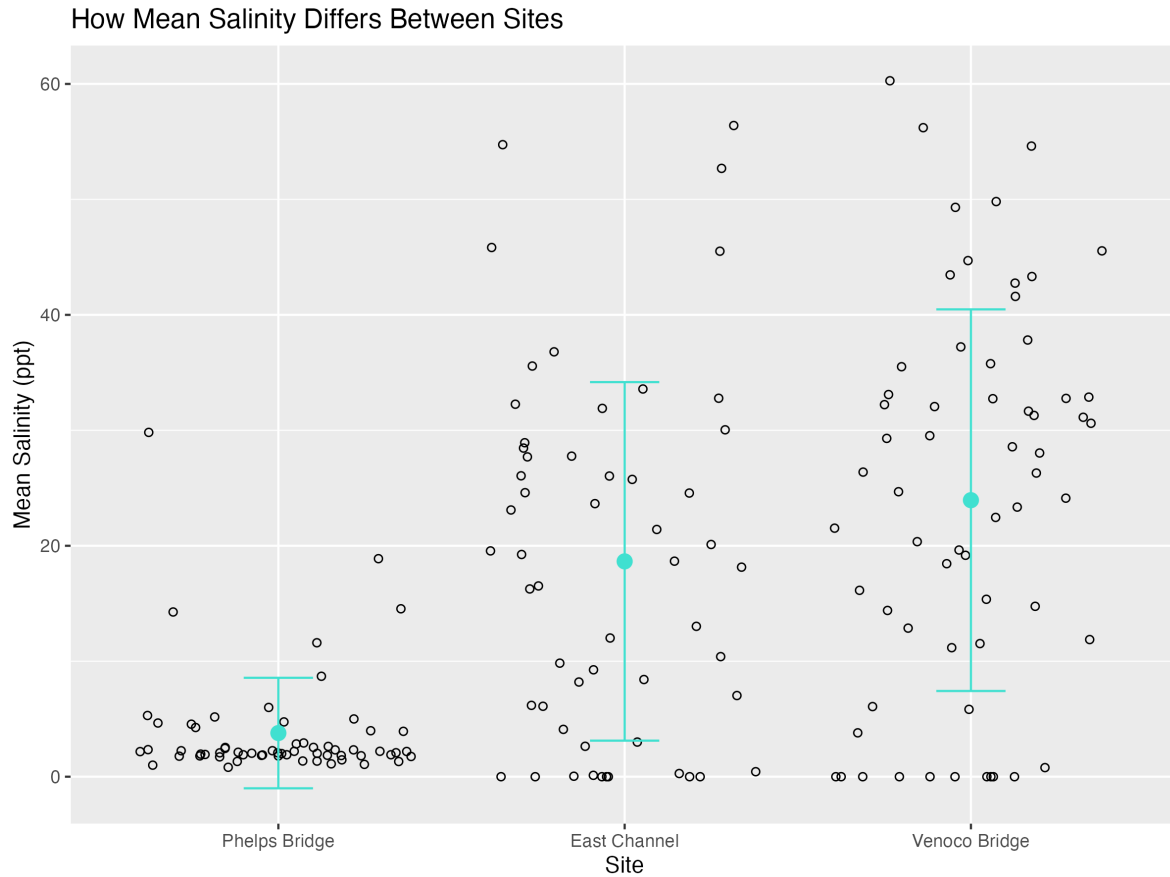


- The addition of this plot was taken as a suggestion made by An after reviewing our Work Plan/Project Proposal. This figure aims to demonstrate that not only does the elevation code of a site yield a difference in monthly mean salinity (as seen in figure 3), but also that the three NCOS sites also differ in their overall monthly mean salinity values. This plot sets the basic understanding of salinity variety across sites while also adding context to one of the visualizations directly answering our question.

Strip Chart w/ Standard Deviation - How Mean Salinity Differs Between Sites

```
ggplot(data = salinity_prdp,
       mapping = aes(x = site_full,
                     y = mean_salinity)) +
  geom_jitter(shape = 21) +
  stat_summary(fun = mean, geom = "point", size = 3, color = "turquoise") +
  stat_summary(fun.data = mean_sdl, fun.args = list(mult = 1), geom = "errorbar", width = 0.2)
```

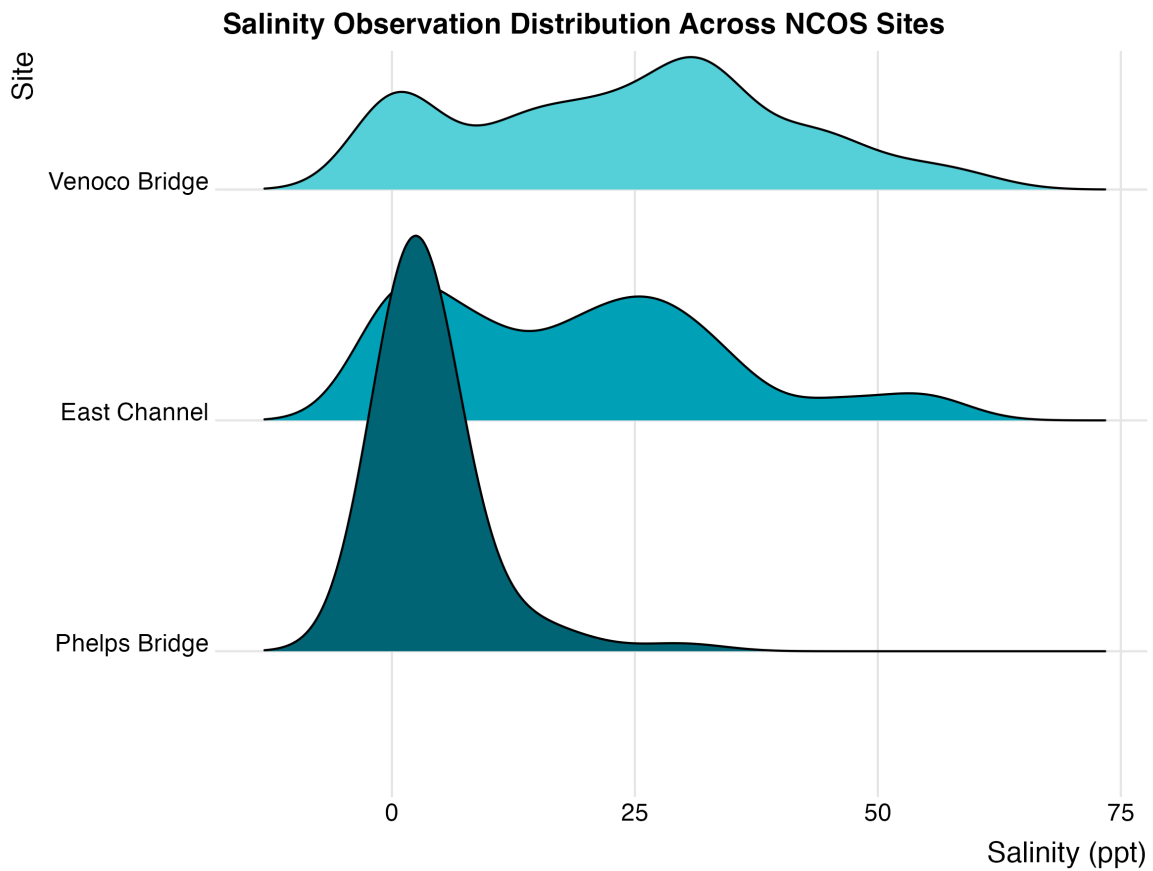
```
labs(x = "Site",
     y = "Mean Salinity (ppt)",
     title = "How Mean Salinity Differs Between Sites")
```



Ridgeline Plot - Salinity Observation Distribution Across NCOS Sites

```
ggplot(salinity_prpc, aes(x = mean_salinity, y = site_full, fill = site_full)) +
  geom_density_ridges() +
  theme_ridges() +
  theme(legend.position = "none") +
  scale_fill_manual(values = natparks.pals("Banff")) +
  labs(x = "Salinity (ppt)",
       y = "Site",
       title = "Salinity Observation Distribution Across NCOS Sites")
```

Picking joint bandwidth of 4.37

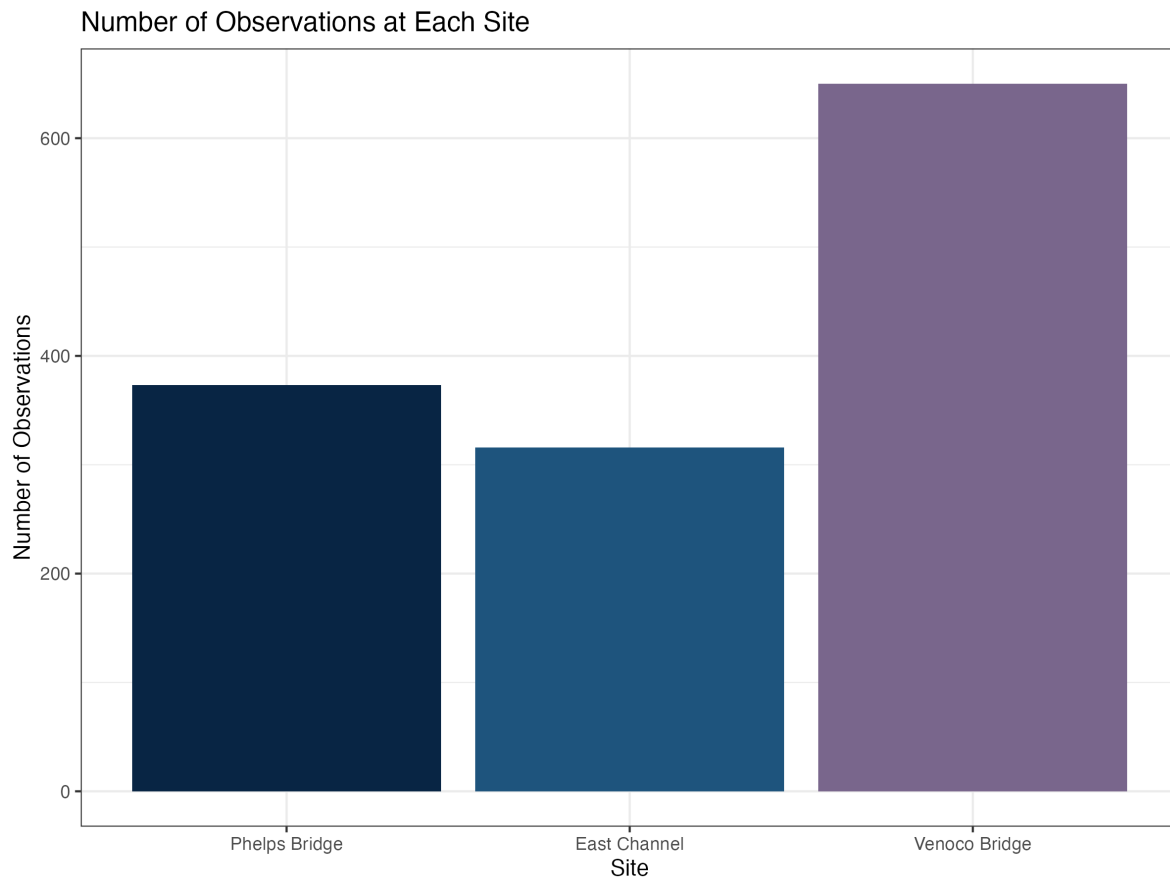


Distribution of Observations

of Observations per Site

```
ggplot(data = water_quality_weather |>
  filter(!global_id %in% salinity_outlier_ids,
    site_name != "other",
    elevation_code %in% c("0", "1", "2"))) |>
  mutate(site_full = fct_relevel(site_full,
    "Phelps Bridge", "East Channel", "Venoco Bridge")),
  mapping = aes(x = site_full)) +
  scale_fill_manual(values = natparks.pals("Volcanoes")) +
  geom_bar(aes(fill = site_full)) +
```

```
theme_bw() +
theme(legend.position = "none") +
labs(x = "Site",
      y = "Number of Observations",
      title = "Number of Observations at Each Site")
```



of Observations per Elevation Code

```
ggplot(data = water_quality_weather |>
  filter(!global_id %in% salinity_outlier_ids,
         site_name != "other",
         elevation_code %in% c("0", "1", "2")),
  mapping = aes(x = elevation_code)) +
geom_bar(aes(fill = elevation_code)) +
```


Observation Count Across Elevation Codes

Elevation Code	Number of Observations
0	485
1	625
2	225

5. Plan for Elective

For our elective, we have chosen to create a hydrology map representing the different levels of salinity throughout

should include some description/image of drafts

should include plan for making progress in the next two weeks